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Deactivation and regeneration of ZSM-5 zeolite in catalytic pyrolysis of plastic wastes

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ABSTRACT

In this work, a study of the regeneration and reuse of ZSM-5 zeolite in the pyrolysis of a plastic mixture has been carried out in a semi-batch reactor at 440 °C. The results have been compared with those obtained with fresh-catalyst and in non-catalytic experiments with the same conditions. The use of fresh catalyst produces a significant change in both the pyrolysis yields and the properties of the liquids and gases obtained. Gases more rich in C3–C4 and H_2 are produced, as well as lower quantities of aromatic liquids if compared with those obtained in thermal decomposition. The authors have proved that after one pyrolysis experiment the zeolite loses quite a lot of its activity, which is reflected in both the yields and the products quality; however, this deactivation was found to be reversible since after regeneration heating at 550 °C in oxygen atmosphere, this catalyst recovered its initial activity, generating similar products and in equivalent proportions as those obtained with fresh catalyst.

1. Introduction

Nowadays, plastic waste management policies are based on energy recovery or elimination processes. In Occidental Europe, about 80 wt.% of the annually generated plastics is landfilled or incinerated (Cimadevila, 2008). However, to fulfil the requirements defined in the European legislation concerning waste management, the application of new recycling processes is an unavoidable task that will have to be carried out in the forthcoming years.

In this sense, "feedstock recycling" includes a very wide range of techniques which offer an opportunity to obtain valuable chemicals from poor quality mixed plastics streams. Broadly speaking, such techniques are focused on producing new non-plastic products from plastic goods, normally monomers or hydrocarbons, and therefore are very different from traditional mechanical recycling processes. Among them, pyrolysis is of the most interest since it is a relatively low cost process from which a broad distribution of products can be obtained. In the pyrolysis process (heating without oxygen), the organic components of the material are decomposed generating liquid and gaseous products, which can be useful as fuels and/or sources of chemicals, while the inorganic material remains unaltered in the solid fraction and free of the binding organic matter, being able to be subsequently recycled.

The use of catalysts confers pyrolysis an additional value since an adequate catalyst can improve the products quality and lead the process towards a good selectivity to more valuable products, even at lower temperatures than in thermal pyrolysis (Buekens and Huang, 1998; Aguado and Serrano, 1999). For this reason, catalytic pyrolysis has been intensively studied in the last years. The catalysts more investigated have been conventional acid solids used in petrochemical feedstocks cracking, as zeolites (e.g., Vasile et al., 2001; Aguado et al., 2009) or FCC catalyst (e.g., Miskolczi et al., 2004; Olazar et al., 2009), as well as non zeolitic mesostructured solids (e.g., Sakata et al., 1999; Aguado et al., 2007). The results reported in the literature show that the cracking ability of the catalysts depends on both their physical (textural properties) and their chemical (acid sites) characteristics. For this reason, one of the most used catalyst is ZSM-5 zeolite, due to its strong acidity and shape selectivity, the latter depending on the particular textural properties (structure and pore size) of each zeolite. These particular properties promote cracking of C–C bonds and determine the chain length of the products obtained (Serrano et al., 2005a; Miskolczi et al., 2006).

However, ZSM-5 zeolite is an expensive catalyst which may condition the economy of the process, due to the high amounts of catalyst which would be necessary in a continuous operating plant. An economic study carried out by Ali et al. (2002) concluded that the catalyst cost (type and amount) is a key factor when it comes to compare the catalytic cracking with thermal cracking technology economics; for this reason, the study of deactivation, regeneration and reuse of catalysts is of the most interest from an industrial implementation point of view.

Up to now, the majority of the studies related to the application of used catalysts in plastic wastes pyrolysis have been carried out with spent FCC catalysts (e.g., Lee et al., 2002; Lin et al., 2010). Some of these studies have shown that the initial activity of such

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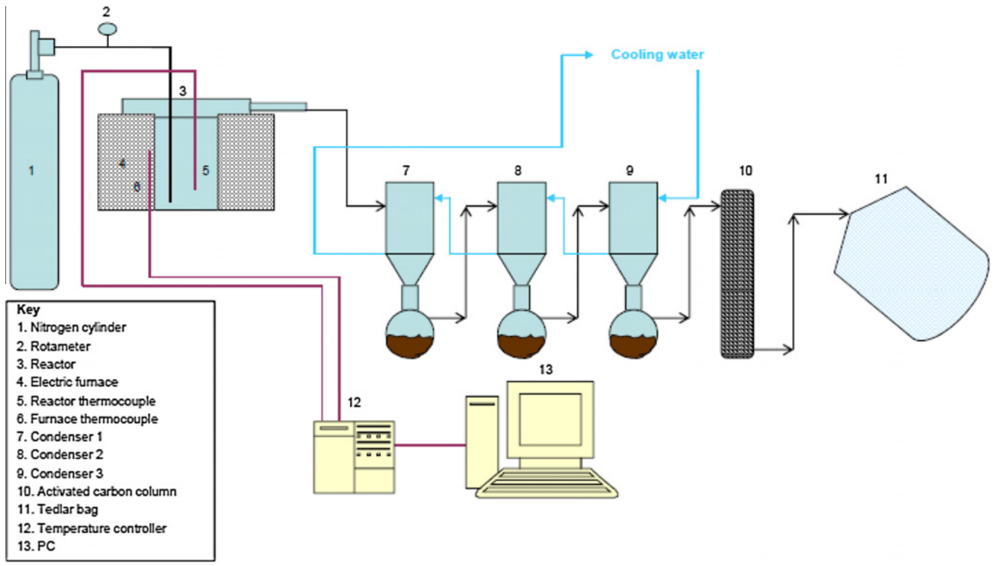

Fig. 1. Flow sheet of the experimental set-up used.

catalysts remains almost unaltered after regeneration and that several regeneration steps can be applied to them without a significant activity loss (Lin and Yang, 2008). There are fewer references in the literature concerning ZSM-5 zeolite regeneration (Serrano et al., 2007; Angyal et al., 2009) and most of them have been carried out in pyrolysis of polyolefins, mainly polyethylene, hindering the problems which can arise due to the interactions among the components when complex plastic mixtures similar to real wastes are pyrolysed.

The objective of this work is to study the loss of activity of ZSM-5 zeolite in order to evaluate if this effective catalyst could be used more than once in pyrolysis of plastic mixtures. With this objective, an exhaustive characterization of the products obtained in thermal pyrolysis and with fresh, regenerated and spent ZSM-5 is presented in this paper.

2. Materials and methods

2.1. Materials

The plastic mixture used for the experiments was composed of 40 wt.% high density polyethylene (HDPE), 35 wt.% polypropylene (PP), 18 wt.% polystyrene (PS), 4 wt.% poly(ethylene terephthalate) (PET) and 3% poly(vinyl chloride) PVC. HDPE, PP and PS were provided by Spanish chemical companies, PVC was provided by a PVC products manufacturing Spanish company, and PET was obtained from a Spanish plastic recycling company. All the plastic materials were used in pellet size (≈3 mm) for the pyrolysis experiments. Additionally finely ground samples (⩽1 mm) were prepared for characterization purposes. The proportions of the plastics in the mixture were established characterizing real samples coming from packing and packaging applications and rejected from an industrial separation plant located in Amorebieta, in the north of Spain (López et al., 2010). The characteristics and composition of the plastic mixture are presented in Table 1. The term "others" in this Table is determined by difference and it includes all elements except carbon, hydrogen, nitrogen and chloride, which are specifically analysed; it must be mainly oxygen coming from PET. As it can be seen, carbon and hydrogen are the main components (84.7 and 12.5 wt.%, respectively) as a consequence of the high HDPE, PP and PS content of the sample (93 wt.%), since these materials are just composed of carbon and hydrogen; this fact also explains its great higher heating value (HHV). It must be mentioned that the sample also contains 1.1 wt.% of chlorine, due to the presence of PVC in the mixture.

A commercial ZSM-5 zeolite provided by Zeolist International was used for the catalytic experiments. The catalyst has a SiO_2/Al_2O_3 ratio of 50 and a total acidity of 0.17 mmol NH_3 g^{-1}; it was used as was received, without any activation operation. In all the experiments, 10 g of zeolite were mixed with the 100 g plastic sample at the beginning of the experiment (liquid phase contact).

2.2. Experimental procedure

The pyrolysis experiments were carried out using a 3.5 dm^3 semi-batch reactor. In a typical run, 100 g of the sample were placed into the reactor which was sealed and heated at a rate of 20 °C min^{-1} up to 440 °C; this temperature was maintained for 30 min, a standard time which was established on the basis of previous studies carried out by the authors with other polymeric wastes in the same installation (Legarreta et al., 1995; de Marco et al., 1995; Torres, 1998). The parameters used in this paper (440 °C, 30 min) are optimized operating conditions as far as energy efficiency, conversion and liquid yields is concerned. These optimized conditions were established by the authors after specific research concerning this topic.

Nitrogen was passed through during the whole experiment at a rate of 1 dm^3 min^{-1} to sweep the decomposition products from the

Table 1
Moisture, ash and elemental composition (wt.%) and HHV (MJ kg^{-1}) of the plastic mixture.

Parameter	Moisture	Ash	C	H	N	Cl	Others[a]	H/C ratio	HHV
Value	0.1	0.0	84.7	12.5	<0.1	1.1	1.5	1.8	43.9

[a] By difference.

reaction medium. The evolved products flowed to a series of running water cooled gas–liquid separators where the vapours were condensed and collected as liquids. The uncondensed products were passed through an activated carbon column and collected as a whole in Tedlar plastic bags. A study about the appropriate configuration for condensation and cleaning of the pyrolysis gases was reported by the authors in a previous paper (de Marco et al., 2009). The optimised experimental set-up is presented in Fig. 1.

The amount of solids (products in the reactor after pyrolysis) and liquids obtained in each pyrolysis run were weighed, and the pyrolysis yields were calculated as weight percentage with respect to the amount of raw material pyrolysed. In the catalytic experiments, the solid yield was calculated in a free catalyst basis. Gas yields were, as a general rule, calculated by difference. Some experiments were specifically devoted to directly quantify the amount of gases by gas chromatography; in such experiments a closure of the mass balance of about 90 wt.% was obtained. The results of the pyrolysis yields which are presented in Section 3 of this paper, are the mean value of at least three pyrolysis runs carried out in the same conditions and which did not differ each other more than three points in the percentage.

2.3. Analytical techniques

Moisture and ash contents of the samples were determined by thermogravimetric analysis according to D3173-85 and D3174-82 ASTM standards, respectively, and the elemental composition with an automatic CHN analyser. Chlorine was determined following the method 5050 of the Environmental Protection Agency (EPA) of the United States. The higher heating value (HHV) was determined with an automatic calorimetric bomb.

Additionally, pyrolysis liquids were also analysed by gas chromatography coupled with mass spectrometry detector (GC–MS). When the match quality of the identification result provided by the MS search engine was lower than 85%, the result was not considered valid and such compounds were classified as "No identified". The compounds names included in the tables of this paper correspond to the tentative assignments provided by the MS search engine and have been contrasted, as far as possible, with bibliographic data and occasionally with calibration standards. Pyrolysis gases were analysed by means of a gas chromatograph coupled with thermal conductivity and flame ionization detectors (GC-TCD/FID). Due to the difficulty in distinguishing among isomers from C3 to C6, such discrimination was not made. The HHV of the gases was calculated according to their composition and to the HHV of the individual components.

The textural properties of the fresh, the regenerated and the used catalyst were determined by means of nitrogen adsorption–desorption isotherms at 77 K in an AUTOSORB-1 Quantachrome equipment. Surface areas were calculated by means of BET equation and external surface areas were obtained applying the *t*-plot method. Total pore volume was measured at $P/P_o = 0.99$. The regeneration of the used catalyst was carried out in two steps in a LECO TGA-500 thermobalance, which is capable of handling at the same time 19 crucibles of 1–5 g capacity each. In the first step the catalyst was heated under nitrogen flow up to 550 °C and maintained there for 30 min and then coke was eliminated by combustion under oxygen flow at the same temperature until no longer weight loss was detected. SEM images were obtained in a LEO 1525 electron microscope and the amount of carbon deposited in the catalyst was determined with the LECO elemental analyser.

3. Results and discussion

3.1. Catalysts characterization

The textural properties of the fresh, regenerated and spent ZSM-5 zeolite are presented in Table 2. With the aim to study the influence of char-coke particles in the spent ZSM-5 characteristics, the textural properties of char obtained with similar conditions in a thermal pyrolysis run have been included in Table 2. BET surface area of ZSM-5 zeolite is 412.0 $m^2\,g^{-1}$, with high micropore area (346.1 $m^2\,g^{-1}$) and also high micropore volume (0.1 $cm^3\,g^{-1}$), which indicates the existence of relatively high internal porosity in this catalyst. The spent ZSM-5, which was characterized after one pyrolysis experiment, showed quite low micropore area (3.0 $m^2\,g^{-1}$). This fact may be explained by the relatively high content of carbon which was deposited in the zeolite surface during pyrolysis (23.0 wt %); this carbon is supposed to be the char which is formed during the process, so it most probably has the same properties as the char of thermal pyrolysis, which are presented in Table 2. The micropore area and micropore volume values for char indicate that char is not a microporous material, and deposited in the zeolite drastically decreases the micropore area of that.

The deposited carbon is also responsible for the higher total pore volume (TPV) obtained in the zeolite after pyrolysis (0.6 $cm^3\,g^{-1}$) compared to the fresh catalyst (0.4 $cm^3\,g^{-1}$), as well as for the noticeable increase in the external surface area (ESA, from 65.9 $m^2\,g^{-1}$ in fresh catalyst to 288.6 $m^2\,g^{-1}$ in spent one). This external area is about 99% of the total surface area (BET, 291.6 $m^2\,g^{-1}$) of the spent ZSM-5, which indicates that almost all micropores have been blocked by coke during the pyrolysis run.

These results are corroborated by the characteristics of the regenerated ZSM-5 zeolite, since after burning the coke (regeneration), the BET surface area of the catalyst was practically the same as that of the fresh one (411.1 and 412.0 $m^2\,g^{-1}$, respectively), and the same occurs with the TPV (0.4 $cm^3\,g^{-1}$ in both samples). Such was not the case of the ESA, which remained higher than in the fresh catalyst (124.3 and 65.9 $m^2\,g^{-1}$, respectively). These data suggest that some changes in the physical structure of the catalyst may have taken place during the regeneration of the zeolite, since the carbon which remains after regeneration (0.7 wt.%) is too low to be responsible for such a high difference in ESA. On the other hand, it cannot be stated that this carbon content corresponds to coke; they could be not removable stable coke species which are formed during pyrolysis of plastic wastes (Serrano et al., 2007), but also carbon from carbonates formed during the regeneration process.

Table 2
Textural properties of the catalysts and of char.

Property	Fresh ZSM-5	Spent ZSM-5	Regenerated ZSM-5	Char
BET surface area ($m^2\,g^{-1}$)	412.0	291.6	411.1	4.7
Ext. surface area ($m^2\,g^{-1}$) (ESA)	65.9	288.6	124.3	4.7
Micropore volume ($cm^3\,g^{-1}$) (MPV)	0.1	1.0E-03	0.1	0.0
Total pore volume ($cm^3\,g^{-1}$) (TPV)	0.4	0.6	0.4	9.5E-03
Micropore area ($m^2\,g^{-1}$)	346.1	3.0	286.8	0.0
Carbon (wt.%)[a]	–	23.0	0.7	94.1

[a] Determined by CHN analysis.

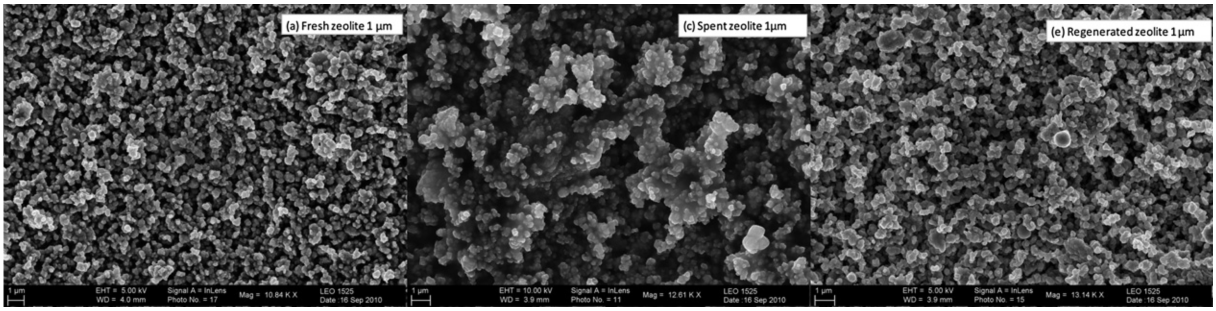

Fig. 2. SEM photographs of the fresh, spent and regenerated zeolite.

SEM photographs of the fresh, spent and regenerated zeolite are presented in Fig. 2 in order to see the effect of deposited coke and regeneration in the surfaces of the catalysts. Fig. 2 shows that fresh ZSM-5 zeolite is formed by uniform crystals with sizes within the nanometer size (≈100–300 nm). Quite higher particles can be observed in the spent zeolite as a consequence of some kind of agglomeration produced by the deposited coke. On the contrary, the SEM photograph of the regenerated zeolite is very similar to that of the fresh one, which indicates that it recovers its textural appearance after the regeneration process.

3.2. Pyrolysis yields

The pyrolysis yields obtained in the thermal and catalytic experiments are shown in Fig. 3. The liquid fraction was the main product in all cases, reaching almost 80 wt.% in the thermal pyrolysis run, followed by gases. The solid fraction, which was practically equivalent in all the experiments (≈3 wt.%), corresponds to char formed during pyrolysis (Grittner et al., 1993; Williams and Williams, 1997; Van Krevelen and Te Nijenhuis, 2009).

The use of fresh ZSM-5 produced a strong increase in the gas yield, which was about 40 wt.% while less than 20 wt.% of gases was obtained in the thermal run. As a consequence fewer liquids were obtained with the fresh catalyst than in the thermal run. This is a well known effect which has been reported before by other research groups (e.g., Miskolczi et al., 2006; Williams and Bagri, 2004) as well as by the authors (de Marco et al., 2009), and it is attributed to both the textural properties of the zeolite and the presence of Brönsted and Lewis acid sites in this catalyst.

After one pyrolysis experiment, the zeolite was recovered and spent directly in the following pyrolysis run; the yields obtained are shown in Fig. 3. These results clearly show that the catalyst has lost its activity, yielding liquids and gases almost in the same proportions as in the thermal run. This quick deactivation has also been observed by other authors and it has been associated with coke deposition, which seems to block the zeolite pores as well as the acid sites (Angyal et al., 2009).

On the contrary, after the regeneration of the zeolite, the yields obtained were very similar to those obtained with the fresh catalyst, showing the effectiveness of the regeneration process and the possibility of using the zeolite more than once if it is properly regenerated. These results are quite in accordance with those obtained by Serrano et al. (2007), who studied the regeneration of two nanocrystalline ZSM-5 catalysts in pyrolysis of PE.

3.3. Pyrolysis liquids

A summary of the GC–MS analyses applied to pyrolysis liquids is shown in Table 3. Aromatic compounds were the main fraction in the liquids, ranging from 67.7% to 97.4% area. It is worth mentioning the aromatization ability of the ZSM-5 zeolite, which produced a great increase in such compounds if compared to the thermal experiment. The capacity of this catalyst to increase aromatics in the liquid fraction has been reported before in the pyrolysis of plastic mixtures (Vasile et al., 2001) as well as in the pyrolysis of pure PE (Serrano et al., 2005b; Marcilla et al., 2009), and it is explained by the presence of high number of Brönsted acid sites (strong acid sites) within this zeolite, where aromatization reactions are favoured. It is also worth noting that the use of ZSM-5 led to the generation of higher quantities of C5–C9 compounds and lower quantities of long chain chemicals (>C13), which is quite in accordance with the well known shape selectivity of this catalyst.

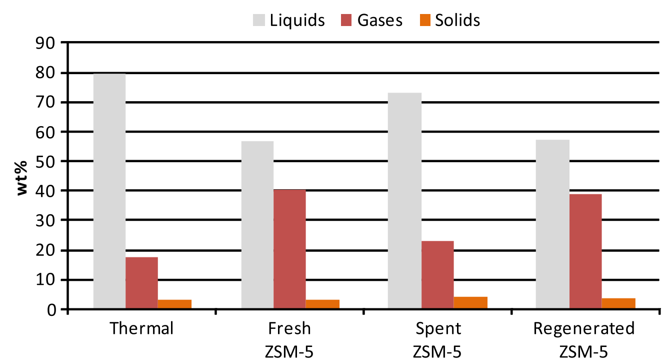

Fig. 3. Pyrolysis yields obtained in the thermal and catalytic experiments.

When the spent ZSM-5 was used, the aromatic content of the liquids drastically decreased from 95.2% to 78.4% area, due to the fact that most of strong acid sites responsible for the aromatization reactions have probably been blocked by coke and char particles; however, the total aromatic content of the liquid obtained in this experiment (78.4% area) was still higher than that obtained in the thermal experiment (67.7% area) which indicates that the zeolite still retains some catalytic activity; this fact can also be observed in the carbon number of liquid fractions, where the decrease in the C5–C9 fraction of the liquids obtained with the spent zeolite is not as high as the mentioned decrease in aromatics. As observed before in the yields discussion, the results obtained with the regenerated ZSM-5 are very similar to those obtained in the fresh catalyst experiment, both in aromatics content and in the carbon number of liquid fractions, indicating that the zeolite has effectively been regenerated.

Table 4 shows the main individual components of the pyrolysis liquids. All of them are aromatic compounds except for dimethylheptene, a branched olefin which has also been obtained by other authors as one of the main products of the cracking of polypropylene (Kaminsky and Zorriqueta, 2007). The products yields confirm the tendency observed in the results presented above, showing similar behaviours for the experiments carried out with fresh and regenerated ZSM-5 and a somewhat intermediate behaviour

Table 3
Aromatic and non-aromatic compounds found in the pyrolysis liquids (% area).

Experiment		Thermal	Fresh ZSM-5	Spent ZSM-5	Regenerated ZSM-5
Total aromatics		67.7	95.2	78.4	97.4
Total non aromatics		22.0	2.7	13.9	1.8
Unidentified		10.3	2.0	7.8	0.8
C5–C9	Aromatics	63.4	78.7	69.4	77.2
	Non aromatics	10.6	2.7	8.1	1.8
	Total	74.1	81.4	77.5	79.0
C10–C13	Aromatics	n.d.[a]	9.5	2.3	12.7
	Non aromatics	5.4	n.d.[a]	5.2	n.d.[a]
	Total	5.4	9.5	7.5	12.7
>C13	Aromatics	4.3	7.5	6.6	6.6
	Non aromatics	5.9	n.d.[a]	0.6	0.6
	Total	10.2	7.5	7.2	7.2

[a] Not detected.

Table 4
Main components of the pyrolysis liquids determined by GC–MS (% area).

Compound	Thermal	Fresh ZSM-5	Spent ZSM-5	Regenerated ZSM-5
Toluene	8.1	12.3	9.9	13.2
Dimethyl-heptene	7.8	1.8	5.4	1.2
Ethyl-benzene	5.7	10.6	9.0	11.2
Xylenes	n.d.[a]	10.1	2.0	13.1
Styrene	46.3	31.4	42.9	23.7
α-Methyl-styrene	3.2	4.5	4.4	3.8
Methyl-napththalene	n.d.[a]	3.3	4.9	4.9

[a] Not detected.

among these ones and the thermal run when spent zeolite was used. Anyway, styrene was found to be the main product in the liquid fraction in all the experiments, followed by toluene and ethylbenzene.

It is somewhat surprising that so much styrene is obtained in the pyrolysis liquids, especially in the thermal and spent ZSM-5 experiments (46.3% and 42.9% area, respectively), since the initial sample is mainly composed of polyolefins (40 wt.% PE and 35 wt.% PP) and only contains 18 wt.% PS. However, previous studies carried out by the authors with the same installation and with plastic samples with similar PS contents, styrene was also the main component of pyrolysis liquids (de Marco et al., 2009; López et al., 2011). The authors have confirmed that such GC–MS peak corresponds to styrene by comparing the mass spectrum of this peak with the theoretical mass spectrums of styrene found in two different databases (Wiley7 and NIST) and by analysing pure styrene (standard quality) with the same conditions in order to compare the result. All the mass spectra (i.e., the databases ones, that of pure styrene and that of the peak assigned to styrene in the GC–MS analysis of the liquids) were almost coincident. This result is almost equivalent to that obtained by Miskolczi et al. (2006), who obtained 53.4% styrene in the pyrolysis liquids of a 90%PE-10%PS mixture at 410 °C.

It is really a difficult task to propose a possible mechanism for styrene formation in such a complex system (five polymeric input materials and many more decomposition products). Concerning thermal decomposition, styrene is formed at a great extent from direct depolymerisation of PS and, as the other aromatic compounds, from secondary reactions among the primary products of the other polymers decomposition, e.g., Diels–Alder type reactions followed by dehydrogenation.

When catalysts are added to the process, it has to bear in mind that both thermal and catalytic decomposition take place at the same time, so both mechanisms are produced simultaneously. In presence of zeolite type catalysts, styrene is formed by thermal mechanisms on the one hand, but on the other hand the "thermally" formed styrene decomposes into other aromatic compounds as a consequence of the secondary reactions that take place to a greater extent in presence of such catalysts. This is the reason why styrene yield is usually lower in catalytic pyrolysis compared to thermal pyrolysis, as it can be seen in Table 4, i.e., the yield of styrene was lower in the fresh and regenerated zeolite runs (31.4% and 23.7% area, respectively) than that obtained in the thermal and spent ZSM-5 experiments (46.3% and 42.9% area, respectively).

This fact is in accordance with the reported by Serrano et al. (2000), who suggested that the lower styrene formation during PS catalytic decomposition with zeolites is produced since the carbenium nature of the acid catalyzed cracking leads to aromatic products different from styrene. The higher yields of toluene, ethyl-benzene and xylenes in the fresh and regenerated catalytic runs may indicate that such compounds are involved in styrene decomposition reactions. Similar behaviour concerning styrene yield was found in thermal and catalytic decomposition over ZSM-5 zeolite by Williams and Bagri (2004) and Miskolczi et al. (2006), when they pyrolysed PS and PS-PE blends, respectively.

The liquids characterization was completed with elemental and HHV analysis; the results are shown in Table 5. The elemental composition of the liquids hardly differ one another, except for slight variations in the carbon and hydrogen contents. These variations are related to the aromatics content of such liquid: the higher the aromatic content, the higher the carbon and the lower the hydrogen content and consequently the H/C ratio, due to the insaturation of the aromatic ring. Taking into account the high aromatics content (67–98% area) it seems, at first, somewhat surprising that in all cases the H/C ratio are around 1.5 instead of close to 1. The explanation to this fact is that H/C = 1 corresponds to non-substituted or lowly-substituted aromatics, but when aromatics are linked to paraffinic or olefinic chains, H/C ratios are higher than 1.

Table 5
Elemental composition (wt.%) and HHV (MJ kg^{-1}) of the pyrolysis liquids.

Parameter	Thermal	Fresh ZSM-5	Spent ZSM-5	Regenerated ZSM-5
C	86.1	87.6	86.4	87.1
H	12.3	11.2	11.6	11.0
Cl	0.2	1.2	0.3	0.6
Others[a]	1.4	0.0	1.7	1.3
H/C ratio	1.7	1.5	1.6	1.5
HHV	44.7	42.2	44.2	42.9

[a] By difference.

It is also important to mention that the HHV of the liquids is very high and similar to that of conventional liquid fuels, showing their potential application as alternative fuels; in this sense, the chlorine contained in these liquids, coming from the PVC of the plastic mixture, could condition their application as liquid fuels. The problem of chlorine may be overcome by stepwise pyrolysis, as it has been demonstrated by the authors in a previous paper (López et al., 2011) which showed that the chlorine content of the liquids coming from PVC containing plastic mixtures can be drastically reduced by carrying out a previous low temperature dechlorination step.

3.4. Pyrolysis gases

The composition of the pyrolysis gases is presented in Table 6. They are composed of hydrocarbons from C1 to C6, carbon dioxide and monoxide and small quantities of hydrogen. The use of fresh catalyst enhanced the production of C3 and C4 fractions compared to the thermal experiment; in the case of spent and regenerated ZSM-5, the production of C4 fraction was even higher than in the fresh zeolite experiment, while the C3 fraction was lower in the former cases. Once more, despite the distribution between C3 and C4 fractions was not the same, the addition of both was similar in the thermal and spent catalyst runs (43.8 and 42.6 wt.%, respectively) and lower than those obtained with fresh and regenerated ZSM-5 (57.0 and 57.1 wt.% respectively), which shows that ZSM-5 promotes the production of such fractions.

It is worth noting that the use of catalysts also increased the production of H_2; this fact has also been reported by other authors (Marcilla et al., 2009; Shah et al., 2010) and may be due to the hydrogen abstraction that takes place during the aromatization reactions, which are favoured by ZSM-5 catalyst.

Table 6
GC-TCD/FID analysis (wt.%) and HHV (MJ kg^{-1}) of pyrolysis gases.

Compound	Thermal	Fresh ZSM-5	Spent ZSM-5	Regenerated ZSM-5
H_2	0.1	0.8	0.7	0.9
CO	0.4	0.7	0.7	0.7
CO_2	2.7	4.9	6.5	4.8
C1–C2				
Methane	9.7	8.8	10.1	9.7
Ethane	12.7	6.5	9.1	6.8
Ethene	8.8	9.3	12.1	9.1
Total	31.2	24.6	31.3	25.6
C3–C4				
C3	26.1	32.4	11.7	21.4
C4	17.7	24.6	30.9	35.7
Total	43.8	57.0	42.6	57.1
C5–C6				
C5	17.4	7.0	13.1	6.7
C6	4.3	4.9	5.1	4.2
Total	21.7	11.9	18.2	10.9
HHV	48.7	46.0	45.8	47.1

Table 7
Moisture (wt.%), elemental composition (wt.%) and HHV (MJ kg^{-1}) of the pyrolysis solids.

Parameter	Thermal	Fresh ZSM-5	Spent ZSM-5	Regenerated ZSM-5
Moisture	n.d.[2]	0.7	0.5	0.6
C	86.6	23.0	31.4	25.9
H	10.0	2.2	2.5	2.6
Cl	<0.1	0.1	0.2	0.1
Others[a]	3.3	74.0	65.4	70.8
H/C ratio	1.4	1.1	1.0	1.2
HHV	42.7	13.1	11.6	14.2

[a] By difference; [2]Not determined.

It has to be mentioned that pyrolysis gases may also contain some chlorine (in HCl form) as it can be deduced from the chlorine mass balance. However, in a potential industrial process such HCl would be absorbed in alkaline solutions by means of wet scrubbers, so that a usable HCl free gas stream would be obtained.

3.5. Pyrolysis solids

Table 7 shows the elemental composition of pyrolysis solids. The great differences among the values of the thermal experiment solid and those of the catalytic ones (specially "others" content) are due to the fact that in the latter the catalysts are mixed with the pyrolysis solids. Concerning the solids of the thermal run, it can be seen that they have quite a high hydrogen content (10 wt.%), which indicates that total decomposition has not been achieved and some unreacted polymer remains in the solid.

There are no significant differences among the solids obtained in the catalytic experiments except for their carbon content. The fresh catalyst showed the lowest carbon content (23.0 wt.%), and the regenerated zeolite a somehow higher carbon content (25.9 wt.%), which indicates that, despite the regeneration process was very effective in terms of yields and products quality, the regeneration process slightly promotes char formation. The highest carbon content was obtained in the solid of the spent ZSM-5 zeolite (31.4 wt.%), quite a high value which justifies the lack of activity of this catalyst due to the strong blockage caused by coke deposition.

4. Conclusions

ZSM-5 zeolite is a very interesting catalyst for pyrolysis of mixed plastic wastes, since high conversions of the samples can be obtained at low temperatures. When it is used, higher conversions than 95 wt.% and liquid fractions with high quantities of valuable aromatic compounds as styrene or toluene are generated, as well as C3–C4 rich gases. The liquid fraction can be used as an alternative to fossil fuels or as a source of valuable chemicals, while the gaseous fraction could be used for power generation for the process itself. Solid yields are lower than 4 wt.%, they are mainly composed of carbon and they can be eliminated from the catalyst by burning them in air.

When using ZSM-5 zeolite in pyrolysis, it is quickly deactivated to a great extent by coke deposition, losing its capacity to produce liquid and gaseous fractions of the same quality of those obtained with the fresh catalyst experiments, although somewhat better quality than that obtained in a thermal run. A regeneration process consisting of burning the deposited coke in an air stream at 550 °C enables the catalyst to recover its initial activity and to produce liquids and gases almost equivalent to those obtained with the fresh catalyst. This is a key factor in the pyrolysis of plastic wastes with ZSM-5 zeolite since having the chance of reusing this catalyst with its whole activity is essential for the economics of a potential industrial pyrolysis process.

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